SDC Hackathon 2026 - Technical & Tools Declaration

Pecel Lele · Adept · unsigned public draft · 31 October 2026. Organiser template sections 1-8 and maximum answer lengths are retained. Source baseline is `main` at `0a7868c` (31 October 2026), not a verified deployment revision. Deadline: 1 November 2026. **Do not submit as a completed accuracy declaration until the leader supplies private fields, checks actual team usage and signs.** No secrets appear here.

1. Team & Project Overview

Organiser field	Response
Team Name	Pecel Lele (confirm registration spelling).
Project / Solution Name	Adept.
Name (Leader)	David Kurniawan (team record; confirm privately).
Email & Phone (Leader)	**PRIVATE COMPLETION REQUIRED** in the organiser's submission copy.
University Affiliation(s)	Xiamen University Malaysia (team record).
Live Prototype URL	https://adept-my.replit.app - previously accessible; final deployment revision and snapshot are unverified here.
Code Repository URL (judge access)	https://github.com/pecel-lelele/adept - confirm final access and merge revision.

2. Architecture & Technology Stack

2.1 System Architecture Overview (maximum 200 words)

Next.js serves a multilingual public interface and proxies requests to private FastAPI. Signed accounts and membership checks protect workspaces. Replit's managed PostgreSQL stores profiles, reference catalogues, snapshots, programmes and progress; local/CI use PostgreSQL with pgvector. The launcher checks API/database readiness; it does not migrate or seed on startup.

Daily GitHub Actions sends an authenticated clock signal to the deployed app. Its shared worker uses its own database connection to retrieve public research, job feeds and course facts, run bounded checks, call the dedicated RunPod MiniLM endpoint for research embeddings, and atomically publish validated dated results. Page reads reuse stored results. Historical fixtures and source outages must be visibly disclosed. Deterministic matching and pathways combine confirmed skills with catalogue records and scenario estimates. Optional Gemini embeddings suggest unfamiliar skills, subject to user confirmation. Optional GLM-5.3 with Gemini failover selects checked Adi answer IDs; server-authored text is displayed.

This is the implemented source architecture, not proof of a production daily refresh. The old GitHub runner could not reach a private Replit database host; the new path needs deployment, required migrations and a successful publication check.

Block flow: public sources → app-side bounded worker (+ RunPod encoder) → validated PostgreSQL snapshot → private FastAPI → Next.js; signed workspaces → deterministic role/pathway/SDG views.

2.2 Languages, Frameworks & Libraries

Source manifest/lock versions, not attested deployed package inventory. Licence terms for binaries and transitive dependencies require independent review.

Name & Version	Purpose in Project	License / terms
Python 3.11; Node.js 24; pnpm 10.29.3	API/runtime, frontend tooling	PSF; Node MIT; pnpm MIT

Next.js 16.4.2; React/React DOM 19.3.1; TypeScript 5.9.3	Public UI, proxy, types	Next/React MIT; TypeScript Apache-2.0
Tailwind CSS and PostCSS 4.1.18; class-variance-authority 0.7.1; clsx 2.1.1; tailwind-merge 3.3.1	Styling and UI variants	MIT except class-variance-authority Apache-2.0
next-intl 4.14.9; Recharts 2.15.4; Three.js 0.180.0; lucide-react ^1.61.0	Languages, charts, mascot, icons	MIT for next-intl/Recharts/Three; ISC for lucide
@fontsource-variable/inter ^5.3.0	Bundled font assets (current UI uses system fallbacks)	Package MIT; Inter font SIL OFL 1.1
openapi-fetch 0.14.1; openapi-typescript 7.9.1	Generated typed API client	MIT
FastAPI >=0.161.1; Pydantic v2, pydantic-settings >=2.16.1; Uvicorn >=0.54	API, validation, ASGI	MIT; Uvicorn BSD-3-Clause
psycopg[binary] >=3.3.6; psycopg-pool >=3.3.3; PostgreSQL 16 + pgvector	Relational/embedding storage	psycopg LGPL-3.0 and bundled notices; PostgreSQL License
PyMuPDF <2; python-multipart >=0.0.32; HTTPX >=0.28.1	PDF/CV parsing, uploads, external requests	PyMuPDF AGPL-3.0 or Artifex commercial; multipart Apache-2.0; HTTPX BSD-3-Clause
RunPod SDK 1.12.0; sentence-transformers 6.2.0; Transformers 5.20.2; PyTorch 2.14.1	Hosted MiniLM worker build/ runtime; deployed image must be checked	RunPod MIT; sentence-transformers/ Transformers Apache-2.0; PyTorch BSD-3-Clause
pytest >=9.1.1, Ruff >=0.17.2, ESLint 9.39.5; uv; Mermaid 11.12.0, PyYAML 6.0.3, actionlint 1.7.7, Prettier ^3.9.10, Playwright	Tests, package/tooling and diagrams	Upstream package terms; predominantly MIT; Playwright Apache-2.0; uv MIT or Apache-2.0

Docker Compose is **local development tooling**, not a cloud host; Docker Desktop terms may differ. Review PyMuPDF's AGPL/commercial terms and transitive notices before attesting licence compliance. Adept's own Apache-2.0 licence does not override them.

3. APIs & Cloud Services

3.1 External APIs

API & Provider	Purpose in Project / execution status	Access Tier
arXiv query API for completed cs.AI weeks; RSS in legacy/manual tooling	Public research observations; historical 2024 replay stays labelled historical	Public/free; terms and rate limits
UN Statistics SDG Goal/Target API	Official definitions and source audits, not endorsement of Adept's links	Public/free
Microsoft Learn Catalog API; Coursera courses API; provider HTTPS pages	Course availability and manually dated duration/fees	Public endpoint/pages; provider terms, not a resale licence
Bank Negara Malaysia OpenAPI	Currency conversion facts for priced learning	Public/free
Greenhouse, Lever, Ashby	Documented employer feeds, location-filtered for Malaysian automatic aggregate counts	Public endpoints; reuse rights not blanket-granted
Himalayas, Jobicy, Remote OK, Remotive; legacy Workday/Jibe	Dated remote postings in the automatic jobs path; legacy historical exports / CLI-only connectors are not the automatic local-count path	Public access with individual provider terms
Careerjet, Jooble, Web3.career	Optional keyed job connectors; historical fetch log marks keys/skips; not needed for automatic path	Keyed; terms and price/activation unverified
OpenDOSM monthly LFS	Malaysian national labour context only, not role-level vacancy demand	Public CC BY 4.0

RunPod Serverless (separate Adi GLM-5.3 endpoint and MiniLM worker)	Optional chat answer selection; batch research embeddings	Sponsor credits apply temporarily; usage billed per applicable service
Google Gemini API (3.5 Flash Lite; 3.6 Flash; 3.1 Flash Lite; embedding-001)	Optional Adi failover, unfamiliar-skill suggestions; trend matcher gated on method equivalence	Free/paid tier depends on account; activation unverified

The **US BLS 2023-2033 baseline** is a committed dated extract, not an API called per user. Model weights are downloaded from Hugging Face during the worker build; the service runs on RunPod. Public reachability does not grant unlimited redistribution rights. No API keys are listed.

3.2 Cloud Platforms & Deployment

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale	Public web and loopback private API at https://adept-my.replit.app	Sponsor-provided Core subscription; screenshot shows Asia / 2 vCPU / 4 GiB at capture time, **not verified today**
Replit integrated PostgreSQL	Separate development/production managed relational databases	Current starting constraint; actual active plan, bill and backup policy unverified
RunPod Serverless and hosted model endpoint	MiniLM worker and optional GLM-5.3 Adi; earlier Pod test was a separate historical workload	Team-recorded US$25 sponsor credits; current region/GPU/bill unknown
GitHub repository, Actions and GHCR	Code, CI, daily clock-trigger workflow, image registry and manual fallback refresh	Public repo / GitHub-hosted runners; private DB host not reachable from ordinary GitHub runners
Local Docker Compose	PostgreSQL/pgvector development and tests only	Local, not deployment

The Replit proof PDF shows historical hosting/database screens and **US$6.38 account spend in a displayed billing window**, not an Adept-only current expense. The RunPod proof PDF shows an earlier 110-skill, 0.354-second Pod vectorization test and Serverless setup; neither proves today's 142-skill catalogue was refreshed or a production trend snapshot published. Do not extrapolate those screenshots to cost per learner.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
all-MiniLM-L6-v2, pinned revision `1110a243fdf4706b3f48f1d95db1a4f5529b4d41`	Sentence Transformers weights on team's RunPod Serverless worker	Pre-trained; no fine-tuning; 384-dimensional public paper/skill embeddings	Apache-2.0
GLM-5.3 (`glm-5.3`)	RunPod-hosted model for optional Adi selection	Pre-trained hosted inference; **selects answer IDs**, not unchecked prose	Hosted provider/model terms; verify exact entitlement
Gemini 3.5 Flash Lite; 3.6 Flash; 3.1 Flash Lite	Google optional failover chain	Pre-trained hosted Adi inference; configured does not establish live invocation	Gemini API Terms
`gemini-embedding-001`	Google optional embedding path	Pre-trained 768-dimensional suggestions and gated trend matcher; no team fine-tuning	Gemini API Terms

Trend method: source arXiv submissions + dated US BLS extract → cleaning/bounded sampling → sentence embeddings/

mention counts → heuristic research velocity and Moderate/Fast scoring → coverage/date/ID validation → stored snapshot. An equivalence report is required before a series switches from MiniLM to Gemini; no cross-method mixing. Research attention is **not** observed Malaysian labour demand, and scenario bands are **not** calibrated confidence intervals.

4.2 Datasets

The counts here are **source CSV rows inspected 31 October 2026**, not a deployed-database count. Taxonomy: **142 skills, 48 stored roles of which 33 `is_target=true`, 104 course rows**; role/skill links are reviewed curations. Historical screenshots and a 20 October synthetic impact fixture list smaller earlier catalogue sizes. No team-trained model or real CV dataset is declared.

Dataset Name	Source / URL	Type	License / rights caveat	Personal Data? (Y/N)
Curated roles/skills, requirements and O*NET 31.0 crosswalk	Team-curated records; O*NET https://www.onetcenter.org/	Team-curated + public-derived	O*NET CC BY 4.0 with exceptions; edited requirements are estimates	N
arXiv cs.AI papers and dated stored mention inputs	arXiv public API https://info.arxiv.org/help/api/	Public metadata / sampled extracts	Metadata/content have distinct rights; raw publications may contain author names	Y (public author names may occur upstream; no learner records)
US BLS 2023-2033 occupation projections	US BLS https://www.bls.gov/emp/tables/occupational-projections-and-characteristics.htm	Public / derived	US government source; US-only baseline, not a Malaysian job forecast	N
Historical 2024 trend replay and synthetic trend fallback	Team-created historical replay from dated arXiv metadata and synthetic fallback	Public-derived / synthetic fixture	Estimated model values, labelled historical/fixture	N
UN 17 goals, 169 targets; team mappings	UN Statistics SDG API https://unstats.un.org/SDGAPI/v1/sdg/Goal/List	Public definitions + team-curated 88 role and 167 skill link rows	UN terms; mappings are team hypotheses, not official approval	N
104 curated courses and prerequisites	Microsoft Learn https://learn.microsoft.com/en-us/training/support/catalog-api ; Coursera https://api.coursera.org/api/courses.v1 ; other cited providers	Public facts + team curation	Content/fees/provider rights remain with providers; claimability must be checked	N
1,175 source job-posting rows; certification and FX extracts	Greenhouse https://developers.greenhouse.io/job-board.html ; Lever https://github.com/lever/postings-api ; Ashby https://developers.ashbyhq.com/docs/public-job-posting-api ; BNM https://api.bnm.gov.my/	Public/curated	Board and cert issuer rights apply; older ads may close	N for learner data; postings may name employers
OpenDOSM national monthly LFS; MyCOL 2024/25 groups	DOSM https://data.gov.my/data-catalogue/lfs_month ; TalentCorp https://www.talentcorp.com.my/resources/publications/mymahir-malaysia-critical-occupations-list-mycol20242025/	Public aggregates / reviewed crosswalk	DOSM CC BY 4.0; TalentCorp terms; historical MyCOL badge does not affect ranking	N
Synthetic demonstration profiles, employers, CVs and regression fixtures	Team-created synthetic examples; no outside source URL	Synthetic	No actual personal CVs used in the competition demonstration	N

4.3 AI-Assisted Development Tools

[x] Yes [ ] No. Team-recorded use (verify each member's usage before the final declaration):

Tool & Version	Used For	Extent

Anthropic Claude Code; model versions varied / not fully recorded	Primary code generation, documentation and presentation planning	Major
OpenAI Codex; exact model versions not fully recorded	Debugging, implementation review and code/document corrections	Major
Replit Agent; version unrecorded	Repository exploration	Minor
Google Antigravity agent; version unrecorded	Repository exploration	Minor
Greptile; version unrecorded	PR review	Moderate
Remotion 4.0.538, ElevenLabs voiceover, Mixkit and Google Fonts	Demo/pitch production only; exact exported Mixkit assets/ font/take credits require team confirmation	Presentation tooling; actual extent to verify

Review and responsibility (maximum 100 words):

The team set Adept's product direction; Claude Code produced much of the source and presentation drafts. Codex helped debug and review; other tools supported inspection and PR feedback. Team members remain responsible for all accepted code and claims. The source is checked against tests, source records and dates; test success does not verify a deployed build or customer benefit. The team must inspect the final media assets, licensing notices and all members' tool usage before signing. Model output never authorises invented catalogue entries or personnel decisions.

The voiceover service is a **presentation service**, not a model inside Adept. The synthesis model and final exported takes remain unrecorded. Do not declare unconfirmed installed plugins as used services.

5. Sponsor-Provided Resources

[x] Yes [ ] No. Team-recorded Replit Core subscription and RunPod **US$25** credits; exact remaining balances and eligibility periods need account confirmation. Sponsorship is not permanent free production hosting.

Sponsor	Resource	How Used
Replit	Core subscription, Autoscale deployment, managed PostgreSQL	Public prototype, private API and persistent app records; Replit Agent exploration declared above.
RunPod	US$25 credits, historical RTX 4090 Pod batch, separate Serverless MiniLM and GLM-5.3 service	Model inference/worker testing; successful Pod batch does not prove new trend publication.

Essential or replaceable (maximum 100 words):

Replit is the prototype's current web/API and managed-PostgreSQL host; RunPod provides MiniLM encoding and optional Adi inference. Neither sponsor's credit is a viable permanent business model. Replacing Replit would require migrating the same schema, signed-session boundaries and backups; replacing MiniLM requires a method-equivalence check before changing any scored series. The learner's normal matching and plan reads are deterministic and do not require per-page model calls. Provider outages preserve dated data or show unavailable states, subject to the actual deployment configuration.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy (maximum 150 words)

The demo uses synthetic identities and CVs, not real applicants. Signed accounts can store names/emails, password hashes, confirmed skills, preferences, memberships, plans, progress and evidence links in PostgreSQL. CV bytes are parsed in memory and not retained. Members explicitly opt in to share skills/progress; personal plans and assistant conversations stay separate. Authentication, server-side membership/manager checks, upload limits, bounded network access and HTTP-only signed cookies mitigate risks.

Optional Gemini suggestions receive short contact-redacted skill phrases. Optional Adi providers receive filtered context and server-authored candidate blocks; redaction cannot guarantee that no personal information leaves the service. MiniLM receives public research/catalogue text. Account export and deletion exist, but automatic retention, tested backup erasure and contractual real-user processing arrangements are **not** established. Do not claim PDPA certification or use real CVs in the demo.

6.2 Known Limitations & Risks (maximum 150 words)

arXiv research share and a US labour baseline cannot measure Malaysian occupation demand. Historical 2024 replay is not fresh evidence; an online check can leave observations unchanged. Scenarios and skill exposure are heuristics, not calibrated probabilities. The catalogue covers selected roles/courses; fees, openings, prerequisites and HRD claimability can change. A completion click is not verified competence or a regulated qualification. SDG links are draft enabling hypotheses, not measured UN outcomes.

Source dates, estimate/fallback labels, ID validation, permissions, bounded workers, stale states and human confirmation mitigate risks; they do not prove universal fairness. A successful model call does not publish data. A local test, public HTTP 200 or sponsor screenshot cannot establish the deployed revision, current observations, vendor licences or customer outcomes. Verify each separately before commercial use.

7. Compliance & Readiness Checklist

Final accuracy attestations require team review; do not check them automatically.

[ ] All actually used APIs, cloud services, models, tools, media and datasets have been checked against each team member's activity.

[ ] Upstream code/data/model licences, notices and presentation asset credits have been reviewed; PyMuPDF distribution terms resolved.

[ ] The final repository, prototype and PDF have been checked for credentials without exposing them.

[ ] AI development assistance has been disclosed and final generated content reviewed.

[ ] Replit/RunPod credits, durations and any additional official sponsor resources have been verified.

[ ] Final URL, deployment revision and judge repository access have been verified.

[ ] Leader's private fields are complete; every limited response passes the word count and the organiser template is complete.

8. Declaration

Unsigned draft. The leader must verify accuracy, complete their contact details in a private copy, and sign/date the organiser declaration. **Signature: [leader to sign]. Date: [actual signing date].** Final required filename: `SDC2026_Pecel-Lele_TechDeclaration.pdf` if this matches the registered team-name convention.